Cross-Latent Velocity-Field Transport for On-Policy Distillation

A Problem Formalization

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1 Problem Setup

Two flow-matching generative models. Let $\mathcal{X} \subseteq \mathbb{R}^D$ be a shared pixel (image) space, and let there be two latent spaces induced by their respective variational autoencoders (VAEs),

$$\mathcal{Z}_A \subseteq \mathbb{R}^{d_A}, \qquad \mathcal{Z}_B \subseteq \mathbb{R}^{d_B},$$

equipped with encoder/decoder pairs $(\mathcal{E}_A, \mathcal{D}_A)$ and $(\mathcal{E}_B, \mathcal{D}_B)$, where $\mathcal{E}_\bullet : \mathcal{X} \to \mathcal{Z}_\bullet$ and $\mathcal{D}_\bullet : \mathcal{Z}_\bullet \to \mathcal{X}$. In general the two latent spaces are *heterogeneous*:

$$d_A \neq d_B, \quad \mathcal{Z}_A \ncong \mathcal{Z}_B, \quad \mathcal{E}_A \neq \mathcal{E}_B,$$

i.e. the channel counts, spatial downsampling rates, and per-channel statistics may all differ.

On each latent space there is a (flow-matching / diffusion) generative model described by a time-dependent velocity field:

$$v_A : \mathcal{Z}_A \times [0,1] \to \mathbb{R}^{d_A}, \qquad v_B : \mathcal{Z}_B \times [0,1] \to \mathbb{R}^{d_B}.$$

Each defines a probability-flow ODE that transports the prior at the noise end $t = 1$ to the latent marginal at the data end $t = 0$:

$$\dot{z}_A(t) = v_A\big(z_A(t), t\big), \qquad \dot{z}_B(t) = v_B\big(z_B(t), t\big). \tag{1}$$

Denote the corresponding latent marginals by $p_A(\cdot, t)$ and $p_B(\cdot, t)$; the clean-end ($t = 0$) marginals are $p_A^0 = \mathcal{E}_A \# p_{\text{data}}$ and $p_B^0 = \mathcal{E}_B \# p_{\text{data}}$. After decoding, both generate the same pixel distribution p_{data}.

Distillation direction. Fix one model as the *teacher* (denoted A, with frozen velocity v_A) and the other as the *student* (denoted B, with trainable velocity $v_B = v_\theta$, parameters θ). The goal of distillation is to use the teacher's generative knowledge in $\mathcal{Z}_A$ to supervise the student's generative process in $\mathcal{Z}_B$.

2 Latent Transport Map

Because v_A and v_B are defined on *different* latent spaces, every teacher signal (clean samples, velocities, transition distributions) lives in $\mathcal{Z}_A$, whereas the student's rollout, loss, and gradients all live in $\mathcal{Z}_B$. Moving the supervision signal across requires a bidirectional map between the latent spaces.

Definition 1 (Latent transport map)**.** A latent transport map is a pair of functions

$$\mathsf{T} : \mathcal{Z}_A \to \mathcal{Z}_B, \qquad \mathsf{T}^{-1} : \mathcal{Z}_B \to \mathcal{Z}_A,$$

satisfying the mutual-inverse relations $\mathsf{T}^{-1} \circ \mathsf{T} = \mathrm{id}_{\mathcal{Z}_A}$ and $\mathsf{T} \circ \mathsf{T}^{-1} = \mathrm{id}_{\mathcal{Z}_B}$ (when the dimensions differ, $d_A \neq d_B$, invertibility can only be required on a common subspace / range, i.e. as a generalized inverse).

Definition 2 (Semantic consistency / pixel anchoring)**.** We call T *pixel-consistent* if it preserves the decoded image semantics: for a teacher latent $z \in \mathcal{Z}_A$,

$$\mathcal{D}_B\big(\mathsf{T}(z)\big) \;\approx\; \mathcal{D}_A(z), \qquad \forall z \in \mathcal{Z}_A. \tag{2}$$

That is, z and $\mathsf{T}(z)$ "represent the same image". This is the semantic-correctness criterion for any reasonable transport.

Definition 3 (Transport consistency error)**.** Define the round-trip pixel-space reconstruction error of the transport,

$$\delta_{\mathsf{T}} \;\coloneqq\; \mathbb{E}_{z\sim p_A}\, \big\|\mathcal{D}_B\big(\mathsf{T}(z)\big) - \mathcal{D}_A(z)\big\|\,, \tag{3}$$

where p_A is the teacher's latent distribution at the relevant time. $\delta_{\mathsf{T}} = 0$ means exact pixel consistency (2). δ_{T} is the scalar that quantifies the lossiness of a transport.

3 Pushforward and Pullback of the Velocity Field

Distillation must transport not only clean samples but the *entire velocity field*, so that the teacher's probability flow in $\mathcal{Z}_A$, after being pushed through T, agrees with the dynamics the student sees in $\mathcal{Z}_B$.

Problem 1 (Transport of the velocity field)**.** Find $(\mathsf{T}, \mathsf{T}^{-1})$ such that the teacher's probability-flow ODE (1) is faithfully transported under the map $z_B(t) = \mathsf{T}\big(z_A(t)\big)$.

Formally, if T is a C^1 diffeomorphism with Jacobian $\mathbf{J}_{\mathsf{T}}(z) \coloneqq \partial\mathsf{T}/\partial z$, the chain rule gives $\dot{z}_B = \mathbf{J}_{\mathsf{T}}(z_A)\,\dot{z}_A$. Motivated by this, we *define* the *pushforward velocity field* (the transported field expressed in z_B coordinates) by

$$(\mathsf{T}\#v_A)(z_B, t) \;\coloneqq\; \mathbf{J}_{\mathsf{T}}\big(\mathsf{T}^{-1}(z_B)\big)\, v_A\big(\mathsf{T}^{-1}(z_B), t\big), \qquad z_B \in \mathcal{Z}_B. \tag{4}$$

The chain-rule identity above is the derivation that justifies this as the correct definition. Symmetrically, mapping a state in the student space back into the teacher space requires the inverse T^{-1} (pullback). Equation (4) makes explicit that, to evaluate the teacher's "opinion" at a state visited by the student, one must simultaneously have:

1. **Invertibility**: access to $\mathsf{T}^{-1}(z_B)$, so the teacher can be queried at the student state z_B;
2. **Differentiability**: the Jacobian $\mathbf{J}_{\mathsf{T}}$ exists and is computable, so the teacher velocity can be correctly mapped into $\mathcal{Z}_B$.

4 Dimension Mismatch and Conditional Pullback

Dimension (mis)match in practice. Empirically, the teacher's VAE has *at least* as many channels as the student's VAE, i.e.

$$d_A \geq d_B,$$

and strictly more in the common case. Two regimes arise:

- *Strict mismatch* ($d_A > d_B$): the forward transport $\mathsf{T} : \mathcal{Z}_A \to \mathcal{Z}_B$ is a *dimensionality reduction* (many-to-one), while the pullback $\mathsf{T}^{-1} : \mathcal{Z}_B \to \mathcal{Z}_A$ is a *dimensional expansion*; recovering a high-dimensional vector from a lower-dimensional one is *underdetermined*, so the student $\to$ teacher transport is **necessarily lossy**.
- *Equal dimension* ($d_A = d_B$): the dimension count alone no longer forces loss; if T is full-rank (invertible) the single-valued pullback is exact, and the constructions below degenerate gracefully to the plain inverse of §2 (made precise in Rem. 1).

This section treats the general case $d_A \geq d_B$, makes the strict-mismatch lossiness precise, and formalizes the operation of "conditioning on the student DiT's hidden states".

Definition 4 (Fiber and ideal pullback)**.** For $z_B \in \mathcal{Z}_B$, the *fiber* (preimage) of the forward transport is

$$\mathcal{F}(z_B) \;\coloneqq\; \mathsf{T}^{-1}(z_B) \;=\; \{\, z \in \mathcal{Z}_A :\; \mathsf{T}(z) = z_B \,\}.$$

When T is a full-rank submersion, a generic fiber is a $(d_A - d_B)$-dimensional submanifold of $\mathcal{Z}_A$ (an affine subspace when T is affine); it has *positive* dimension exactly when $d_A > d_B$, and collapses to a *single point* when $d_A = d_B$. Any *single-valued* pullback $\mathsf{T}^{-1} : \mathcal{Z}_B \to \mathcal{Z}_A$ must select a *unique* representative on each fiber, thereby discarding the in-fiber coordinate—a vacuous choice when the fiber is a point ($d_A = d_B$), and a genuinely lossy one when it is positive-dimensional ($d_A > d_B$). Moreover, let the teacher latent that is pixel-consistent (Def. 2) with z_B be the *ideal pullback*

$$\bar{z}_A(z_B) \;\coloneqq\; \mathcal{E}_A\big(\mathcal{D}_B(z_B)\big) \;\in\; \mathcal{Z}_A, \tag{5}$$

the semantically correct target that the pullback should reproduce.

Problem 2 (Irreducible loss of the underdetermined pullback)**.** Suppose a single-valued pullback T^{-1} has range contained in some d_B-dimensional subset $\mathcal{R} \subseteq \mathcal{Z}_A$ (e.g. a linear pullback $\mathsf{T}^{-1}(z_B) = A^{+}(z_B - b)$ lands in the column space of A^{+}), and let $\Pi_{\mathcal{R}}$, $\Pi_{\mathcal{R}}^{\perp}$ be the corresponding orthogonal projectors. Then for *any* pullback that depends on z_B alone,

$$\mathbb{E}_{z_B}\,\Big\|\bar{z}_A(z_B) - \mathsf{T}^{-1}(z_B)\Big\| \;\geq\; \mathbb{E}_{z_B}\,\Big\|\Pi_{\mathcal{R}}^{\perp}\,\bar{z}_A(z_B)\Big\|, \tag{6}$$

i.e. the energy of the ideal pullback lying in the *teacher-exclusive subspace* $\mathcal{R}^{\perp}$ (the extra $d_A - d_B$ dimensions) cannot be recovered from z_B alone. This is the precise meaning of the student $\to$ teacher transport being "underdetermined and lossy".

Remark 1 (The equal-dimension boundary $d_A = d_B$)**.** When $d_A = d_B$ and the chosen T is invertible, the recoverable subspace is the whole space, $\mathcal{R} = \mathcal{Z}_A$, hence $\Pi_{\mathcal{R}}^{\perp} = 0$ and the right-hand side of (6) vanishes: a single-valued pullback can be exact and there is *no* dimension-induced loss. The fiber is a point, so $\bar{z}_A(z_B) = \mathsf{T}^{-1}(z_B)$, and the conditional pullback introduced below carries no information-theoretic gain ($I(\bar{z}_A; h_\theta \mid z_B) = 0$, since $\bar{z}_A$ is then a deterministic function of z_B); requirement (P4) thus collapses to (P2). This concerns only the *dimension-induced* underdetermination: a residual approximation error may still persist if the chosen transport class (e.g. affine) cannot represent $\bar{z}_A$, but that error is measured by δ_{T} (Def. 3), not by (6). In short, $d_A = d_B$ is the degenerate boundary of this section at which everything reduces to §2–§3.

Conditional pullback: injecting information via the student DiT's hidden states. While computing $v_\theta(z_B, t, c)$ (with c the external conditioning, e.g. the text prompt), the student transformer produces intermediate hidden states

$$h_\theta(z_B, t, c) \;\in\; \mathcal{H},$$

whose dimension is typically far larger than d_B. Augmenting the pullback's domain from $\mathcal{Z}_B$ to $\mathcal{Z}_B \times \mathcal{H}$ yields the *conditional pullback*

$$\mathsf{T}_c^{-1} :\; \mathcal{Z}_B \times \mathcal{H} \to \mathcal{Z}_A, \qquad \hat{z}_A \;\coloneqq\; \mathsf{T}_c^{-1}\big(z_B,\; h_\theta(z_B, t, c)\big). \tag{7}$$

Remark 2 (Essence: turning an underdetermined inversion into conditional inference)**.** The essence of this operation is to use the student's own internal representation as *side information* to estimate the missing coordinate in the teacher-exclusive subspace $\mathcal{R}^{\perp}$, thereby rewriting an "ill-posed dimensional-expansion inversion" as a "conditional inference / conditional completion" problem. Its attainable accuracy is governed by the *conditional entropy*:

$$H\big(\bar{z}_A \,\big|\, z_B,\; h_\theta\big) \;\leq\; H\big(\bar{z}_A \,\big|\, z_B\big), \qquad \text{gain} \;=\; I\big(\bar{z}_A;\; h_\theta \,\big|\, z_B\big) \;\geq\; 0. \tag{8}$$

Under squared error, the optimal estimator is the conditional expectation $\hat{z}_A^\star = \mathbb{E}[\bar{z}_A \mid z_B, h_\theta]$, whose irreducible residual is the conditional variance $\mathbb{E}\operatorname{Var}\big(\Pi_{\mathcal{R}}^{\perp}\bar{z}_A \mid z_B, h_\theta\big)$.

Remark 3 (A necessary caveat: hidden states do not create information from nothing)**.** Since $h_\theta(z_B, t, c)$ is a *deterministic* function of (z_B, t, c), the genuine information gain from conditioning can *only* come from inputs beyond z_B, namely the time t and the external conditioning c:

$$I\big(\bar{z}_A;\ h_\theta \,\big|\, z_B\big) \ \le\ I\big(\bar{z}_A;\ (t,c) \,\big|\, z_B\big),$$

which is strictly positive iff the teacher-exclusive coordinate $\Pi_{\mathcal{R}}^{\perp}\bar{z}_A$ is correlated with (t, c). The role of the hidden states is therefore twofold: (i) they are the natural carrier of the (z_B, t, c) context (in particular the prompt c, which is *not* a function of z_B); and (ii) they are a precomputed, rich nonlinear feature that makes the conditional completion learnable in practice. They *cannot* recover the part of the information that was *physically deleted* by T in the forward direction and is uncorrelated with (t, c).

Remark 4 (Compatibility with the pushforward)**.** Substituting the conditional pullback into the pushforward (4):

$$(\mathsf{T}\#v_A)(z_B, t) \ =\ \mathbf{J}_{\mathsf{T}}(\hat{z}_A)\, v_A(\hat{z}_A, t), \qquad \hat{z}_A = \mathsf{T}_c^{-1}\big(z_B, h_\theta(z_B, t, c)\big).$$

For an affine $\mathsf{T} = Az + b$, the Jacobian $\mathbf{J}_{\mathsf{T}} \equiv A$ is constant, so conditioning *does not alter* the velocity-transport operator A; it only *relocates* the teacher query point $\hat{z}_A$ to a more correct position within the fiber $\mathcal{F}(z_B)$. That is, it sharpens *which* teacher opinion is queried without breaking the linear structure of the transport.

5 Formal Requirements of On-Policy Distillation

The defining feature of on-policy distillation is that the supervision signal must be evaluated on *the state distribution that the student itself rolls out*, rather than on states presampled by the teacher.

The student's on-policy states. The student rolls out a trajectory $\{z_B(t)\}_t \subset \mathcal{Z}_B$ using its own velocity v_θ, inducing a distribution ρ_θ of visited states (it depends on the current parameters θ, hence "on-policy").

The supervision signal. For each visited state $z_B \sim \rho_\theta$, distillation requires the teacher's "opinion" on how to denoise at that point. But the teacher velocity v_A is defined only on $\mathcal{Z}_A$, so its legitimate value in $\mathcal{Z}_B$ is exactly the pushforward field $(\mathsf{T}\#v_A)(z_B, t)$ of (4), which depends on T^{-1} (pulling z_B back to $\mathcal{Z}_A$ to query the teacher) and $\mathbf{J}_{\mathsf{T}}$ (mapping the result back into $\mathcal{Z}_B$).

Distillation objective (general form). On-policy distillation minimizes the discrepancy between the student velocity and the transported teacher velocity over the student's visited distribution:

$$\min_\theta\ \mathbb{E}_t\, \mathbb{E}_{z_B\sim\rho_\theta}\ \Big[w(t)\, \mathcal{M}\Big(v_\theta(z_B, t),\ (\mathsf{T}\#v_A)(z_B, t)\Big)\Big], \tag{9}$$

where $w(t)$ is a time weight and $\mathcal{M}(\cdot,\cdot)$ is a discrepancy measure (e.g. a quadratic regression error, or a divergence between the transition distributions induced by the dynamics); its concrete form is outside the scope of this note. The objective is well defined precisely when $(\mathsf{T}\#v_A)$ can be evaluated on the support of ρ_θ, i.e. when T^{-1} and $\mathbf{J}_{\mathsf{T}}$ exist and are computable.

6 Formal Problem Statement

Combining §1–§5, cross-latent on-policy distillation reduces to the following transport-map design problem.

Problem 3 (Cross-latent transport design)**.** Given two heterogeneous latent spaces $\mathcal{Z}_A, \mathcal{Z}_B$, their decoders $\mathcal{D}_A, \mathcal{D}_B$, the teacher velocity v_A, and the student velocity v_θ, find a pair of maps $(\mathsf{T}, \mathsf{T}^{-1})$, $\mathsf{T} : \mathcal{Z}_A \to \mathcal{Z}_B$ and $\mathsf{T}^{-1} : \mathcal{Z}_B \to \mathcal{Z}_A$, such that:

(P1) **Semantic consistency**: T is pixel-consistent, i.e. the transport consistency error δ_{T} (Def. 3) is as small as possible, $\mathcal{D}_B(\mathsf{T}(z)) \approx \mathcal{D}_A(z)$;

(P2) **Invertibility**: T^{-1} satisfies the mutual-inverse relations of Def. 1, so student states can be pulled back to $\mathcal{Z}_A$ to query the teacher;

(P3) **Differentiable pushforward**: T is differentiable and its Jacobian $\mathbf{J}_{\mathsf{T}}$ is available, so the pushforward velocity field (4) $(\mathsf{T}\#v_A)$ can be evaluated at the student-visited states;

(P4) **Conditional invertibility under mismatch** (active only for $d_A > d_B$): with $d_A \geq d_B$, the loss of a single-valued pullback $\mathsf{T}^{-1}(z_B)$ is lower-bounded by (6). When $d_A > d_B$ this bound is generally positive (underdetermined expansion, necessarily lossy), and one should use the conditional pullback $\mathsf{T}_c^{-1}\big(z_B, h_\theta(z_B, t, c)\big)$ (§4), using the student DiT's hidden states as side information to complete the teacher-exclusive subspace $\mathcal{R}^\perp$, so the pullback error approaches the conditional-variance lower bound (8). When $d_A = d_B$ the bound vanishes and this requirement reduces to (P2) (Rem. 1).

When (P1)–(P4) hold, the supervision signal of the on-policy distillation objective (9) is type-legal and semantically correct, and the distillation problem becomes well posed.

Remark 5 (Hierarchy of requirements)**.** Different strengths of distillation impose different requirements on $(\mathsf{T}, \mathsf{T}^{-1})$: performing only *sample transport* $z_B^0 := \mathsf{T}(z_A^0)$ on clean samples z_A^0 requires only (P1), whereas performing *velocity-field transport* on the student's on-policy trajectory additionally requires (P2)–(P3), and, under *strict* dimension mismatch $d_A > d_B$, (P4) (at $d_A = d_B$, (P4) reduces to (P2)). These levels constitute the several strength variants of Problem 3.

Remark 6 (Scope statement)**.** This note gives only the problem formalization (setup, required properties, problem statement). It does not discuss any concrete construction, parameterization, initialization, or solution algorithm for $(\mathsf{T}, \mathsf{T}^{-1})$, nor does it compare or select among candidate methods.

Remark 7 (External concrete T^{-1}: Cross-Space Bridge)**.** Cross-Space Distillation (arXiv:2606.32020) learns a Bridge $\mathcal{B}_\phi : \mathcal{Z}_B \to \mathcal{Z}_A$ (Student-decoder spatial prior + projector) so that distribution-based one-step distillation (VSD/GAN) can run under Cross-Resolution and Cross-VAE mismatch. In our terms this is a concrete T^{-1} satisfying a soft version of (P1)–(P2) for *sample* transport into Teacher space; it does not by itself supply the Jacobian pushforward (P3) needed for Student-space velocity matching. See `docs/xopd/cross_vae/cross_space_distillation_bridge.tex` for the mapping onto Routes A–C and the XOPD L0/L1 hierarchy (Rem. 5).